

Question 1:

Question: A company is considering two financing options: Option A involves issuing \$1,000,000 in bonds with a 5% coupon rate, and Option B involves issuing 100,000 shares of common stock at \$10 per share. The company expects to generate \$200,000 in annual earnings before interest and taxes (EBIT). Assuming a corporate tax rate of 30%, which financing option will result in higher earnings per share (EPS), and by how much?

Type: short-answer

Options: No options provided.

Knowledge points:

- Debt vs. equity financing impact on EPS
- Tax shield benefits of debt
- Calculating earnings per share under different capital structures

Question 2:

Question: A corporation is considering a new investment project that is expected to generate significant long-term profits but will reduce short-term earnings and, consequently, executive bonuses this year. The CEO, whose compensation is heavily tied to annual performance metrics, is hesitant to approve the project. What term best describes this conflict of interest?

Type: multiple-choice

Options:

A. Double taxation

- B. General partnership liability
- C. Agency problem
- D. Limited partner role
- E. Corporate bylaws

****Knowledge points:**** Agency problem, CEO compensation structure, Short-term vs. long-term decision-making in corporations.

Question 3:

****Question:**** A bond is currently selling for \$950, has a face value of \$1,000, and pays an annual coupon of 6%. The bond has 10 years remaining until maturity. Which of the following statements is true about this bond?

****Type:**** multiple-choice

****Options:****

- A. The bond is selling at a premium, and its yield to maturity is less than the coupon rate.
- B. The bond is selling at a discount, and its yield to maturity is greater than the coupon rate.
- C. The bond's current yield is equal to its yield to maturity.
- D. The bond's price will increase if market interest rates rise.
- E. The bond's coupon payments are not affected by changes in market interest rates.

****Knowledge points:****

- Bond price and face value relationship (discount vs. premium)
- Yield to maturity vs. coupon rate
- Impact of interest rate changes on bond prices
- Current yield vs. yield to maturity

Question 4:

Question: A bond with a duration of 8 years is compared to another bond with a duration of 4 years. Assuming all else is equal, which of the following statements is true regarding their interest rate risk?

Type: multiple-choice

Options:

- A. The bond with 8-year duration has higher interest rate risk.
- B. The bond with 4-year duration has higher interest rate risk.
- C. Both bonds have the same interest rate risk because their coupon rates are equal.
- D. Interest rate risk cannot be determined without knowing the bonds' yields to maturity.
- E. The bond with the higher coupon rate will always have lower interest rate risk.

Knowledge points: Bond duration and interest rate risk, Impact of duration on price sensitivity, Role of coupon payments in duration.

Question 5:

Question: A bank offers two savings accounts with the following interest rates:

- **Account X:** APR of 5%, compounded monthly.
- **Account Y:** APR of 5.1%, compounded annually.

Assuming you plan to invest \$10,000 for five years and will not make any withdrawals, which account should you choose and why?

Type: multiple-choice

Options:

- A. **X**, the effective annual rate is higher.
- B. **X**, the total interest earned will be lower.
- C. **Y**, the annual percentage rate is higher.
- D. **Y**, the effective annual rate is higher.
- E. **Both accounts will yield the same future value.**

Knowledge points:

- Effective Annual Rate (EAR) calculation
- Impact of compounding frequency on returns
- Comparison of different APR structures

Question 6:

Question:

ABC Corporation just paid a dividend of \$2.50 per share. The dividends are expected to grow at a constant rate of 5% per year indefinitely. If the required rate of return for the stock is 9%, what is the current stock price? Additionally, what is the dividend yield and the capital gains yield for ABC Corporation?

Type: short-answer

Options: No options provided.

Knowledge Points:

Dividend Discount Model (DDM) for constant growth stocks, Required return calculation, Dividend yield and capital gains yield relationship.

Question 7:

Question: A corporate bond has a face value of \$1,000, matures in 15 years, and pays a 5%

coupon rate semi-annually. If the bond is currently priced at \$950, what is the yield to maturity

(YTM)? Type: multiple-choice Options:

- A. 5.25 percent
- B. 5.50 percent
- C. 5.75 percent
- D. 6.00 percent
- E. 6.25 percent

knowledge points: Bond pricing and YTM relationship, Coupon rate calculation, Semi-annual coupon payments

Question 8:

****Question:****

Five years ago, TechGrow Inc. paid a dividend of \$5 per share. The company has since adopted a policy where dividends will grow at a rate of 6% per year indefinitely. The required rate of return for TechGrow's stock is 11%.

- a. What is the current stock price if the next dividend (D_1) is to be paid one year from today?
- b. TechGrow also issued preferred stock with an annual dividend of \$7, paid quarterly (i.e., \$1.75 per quarter). If the next dividend is due in three months, what is the value of the preferred stock today?
- c. A competitor, InnovateCorp, paid a dividend of \$4 per share yesterday. Analysts predict that InnovateCorp will experience a high growth rate of 15% per year for the next five years, after which the growth rate will stabilize at the same rate as TechGrow (6%). Assuming a discount rate of 11%, what is the current value of InnovateCorp's stock?

Type: short-answer

Options: No options provided.

Knowledge points:

- Dividend Discount Model with constant growth
- Preferred stock valuation with quarterly dividends
- Two-stage Dividend Discount Model for high and stable growth

Question 9:

New Question:

You are planning to buy a car and have two financing options:

Option 1: Take a loan of \$25,000 today, repayable in equal monthly installments over 5 years at an annual interest rate of 6% (compounded monthly).

Option 2: Lease the car with monthly payments of \$450 for 5 years, with the first payment due today. At the end of the lease, you can buy the car for a residual value of \$10,000.

Assume the appropriate discount rate for evaluating the lease is 5% per year (compounded monthly).

- What is the monthly loan payment under **Option 1**?
- What is the present value of the total cost of **Option 2** (lease payments + residual value)?
- Which option is more cost-effective based on present value?

Type: long-answer

****Options:**** No options provided.

****Knowledge Points:**** Time Value of Money (TVM), Annuity valuation, Present value of cash flows, Loan amortization.

Question 10:

Question: Which of the following statements about the term structure of interest rates is

correct? Type: multiple-choice Options:

- A. An upward-sloping yield curve implies that investors expect higher inflation in the future.
- B. A downward-sloping yield curve suggests that long-term real interest rates are lower than short-term real rates.
- C. A flat yield curve indicates that the interest rate risk premium is at its highest level.
- D. The expectations theory alone can fully explain all observed yield curve shapes.
- E. A steep yield curve always signals an impending economic recession.

knowledge points: Term structure and inflation expectations, Slope of term structure and interest rate risk premium, Downward sloping term structure and real rate relationship

Question 11:

****Question:****

You are planning to save for your child's college education. Your child will start college in 18 years, and you want to have \$150,000 saved by that time. You plan to make equal annual deposits into a college savings account, starting today and continuing until the day your child starts college (a total of 18 deposits). The account earns an annual interest rate of 6%.

(a) How much do you need to deposit each year to reach your goal?

(b) If you decide to make a single lump-sum deposit today instead of annual deposits, how much would you need to deposit to achieve the same goal?

****Type:**** long-answer

****Options:**** No options provided.

****Knowledge Points:**** Time Value of Money, Present Value Calculations, Future Value of Annuities, Compound Interest Application.

Question 12:

****Question:****

A bond with a face value of \$1,000 pays a semi-annual coupon and has 12 years to maturity. The bond is currently selling for \$1,150 and has a yield to maturity (YTM) of 6.5%.

- a. What is the current yield of the bond?
- b. If the YTM decreases by 1.5% three years from now and you sell the bond at that time, what will be the selling price and the holding period yield (HPY) of your investment?

****Type:**** short-answer

****Options:**** No options provided.

****Knowledge points:**** Current yield calculation, Bond price determination with changing YTM, Holding Period Yield (HPY) calculation.

Question 13:

****Question:**** You are taking out a \$200,000 mortgage loan with a 30-year term at a fixed annual interest rate of 4.5%. The loan requires monthly payments and has a balloon payment due at the end of 10 years.

- (a) Calculate the monthly payment for this loan.
- (b) Determine the remaining loan balance (balloon payment) at the end of 10 years.

(c) If you decide to refinance the remaining balance at the end of 10 years with a new 20-year loan at 4.0% interest, what will be the new monthly payment?

****Type:**** long-answer

****Options:**** No options provided.

****Knowledge Points:****

- Time Value of Money
- Loan Amortization
- Balloon Payment Calculation
- Refinancing and Payment Recalculation

Question 14:

****Question:****

ABC Corporation just paid a dividend of \$3 per share. The dividends are expected to grow at a high rate of 12% per year for the next five years. After that, the growth rate will stabilize at a constant rate of 5% per year indefinitely. If the required return on the stock is 8%, what is the current value of ABC Corporation's common share?

****Type:**** short-answer

****Options:**** No options provided.

****Knowledge points:****

Dividend Discount Model DDM (two-stage growth), Present value of irregular dividends, Required return and growth rate phases.